Desktop: NVIDIA: GeForce - Gaming | AMD: Radeon - Gaming
Workstation: NVIDIA: Quadro - Design; Tesla - Computation | AMD - FirePro; Design and computation

Specifications that matter

Both competitors have some impressive proprietary technologies which bring out all the shiny goodness while playing games to boast of. Though these cards are not well suited for heavy computation loads they are capable of crunching out some great numbers occasionally. So if you are primarily a gamer then the desktop series are what you need to consider.

There are games that favour one particular manufacturer and the games look stunning on them, while on the competitor's card they do look great but the stark difference is clearly noticeable. This in turn doesn't mean that you should end up buying cards from both manufacturers and plug in whichever renders the game best (though there are people who do that).

Now here, greater the number, better the performance unless that parameter is temperature.

Core frequency/base frequency

This is the speed at which the GPU card runs, the higher this number, the more powerful the card will be and subsequently more is the heat that will be generated.

Memory type

Almost all cards today use GDDR5 type memory but there are variants that are rebranded GPUs of previous generations that still use DDR3 memory and are thus, slower.

However, the older GPUs were only capable of handling so much and it's only the newer cards which are capable of utilising the bandwidth that comes with GDDR5.

DirectX and OpenGL support

If you are into playing a lot of games then knowing what version of DirectX is important.

Then again all cards since the past 3 years do support DirectX 11 and you don't need to worry about that. The same goes for OpenGL.

DECISION TIME GPUS

Whether you need to buy a new graphics card comes down to finding out which component happens to be the bottleneck in your system.

And once that is established there are multiple ways to get more performance. Benchmarking web-sites have records of various configurations, all you need to do is check which of the following gets you the most in terms of performance.

1. Overclocking your current card - A little OC, even if it is around 50-100 MHz helps out.
2. Scaling it up - Get the same model that you currently own and link it in an SLI/CrossFireX configuration. This is quite effective given that you may be able to get performance similar to that of high end cards at roughly half the price or even lower. There are certain criteria that need to be kept in mind to enable a proper SLI/CF configuration which you can easily find on the internet.
3. Sell your current card to get the extra cash and get a good card which can last you 2-3 years.

Check out the comparison test in our August 2012 issue for more details. If your card is 4 generations or more behind then grab some money, get out and get a card from the latest family from either competitor.

Monitors supported

By this we don't mean models but simply the number of monitors that can be run off a single card. NVIDIA's 600 series supports 4 monitors while AMD's 6000/7000 series supports 6 monitors.

Technology

If you are into programming and wish to work with a certain technology like CUDA then that must be considered. However, if you use technology which can be run on both NVIDIA and AMD cards then you need not bother.

NVIDIA

CUDA Cores: NVIDIA's unified shader architecture takes care of pretty much all the calculations there are. They are mentioned under the label "Shaders" in GPU-Z.
Boost clock - With the 600 series onwards a new feature called "Boost frequency" was introduced, which is similar to the turbo boost found on processors.

AMD

Shaders - The AMD shader count and NVIDIA shader count varies because they're interpreted differently by them and should not be equated on a performance scale.

Things that don't matter

The brand

Most brands use the reference PCB for their mid-range and low-end cards, so brand isn't a factor. Though, once you start pondering over the premium segment the brand does come into perspective, certain brands use buzzwords to describe common technology while some do put the extra moolah into some R&D and bring out technology worth considering. All brands use great coolers for premium cards in the high-end segment and for the majority of consumers, this isn't that big a factor. Just go with the brand that you trust and provides good after sales service in your region.

Factory overclock

Overclocking low end cards can cause a lot of problems as the coolers aren't designed to handle the extra load and for the mid-range cards, using tools like MSI's Afterburner will help you get similar performance without shelling out extra for a premium card. Premium cards scale well and as in the mid-range segment, can be easily overclocked.

Memory

This one has been done to death, there are very few games for whom massive amounts of Graphics memory is required, for the absolute majority you need not go for extra memory.

Recommendations

High-end (₹30,000 - 48,000)

ASUS GTX 680-DC2T
ZOTAC GTX 680
ASUS GTX 670-DC2T

Mid-range (₹10,000 - 30,000)

ASUS HD7770- DC2T

Low-end (Below ₹10,000)

Galaxy GT 640 1GB.